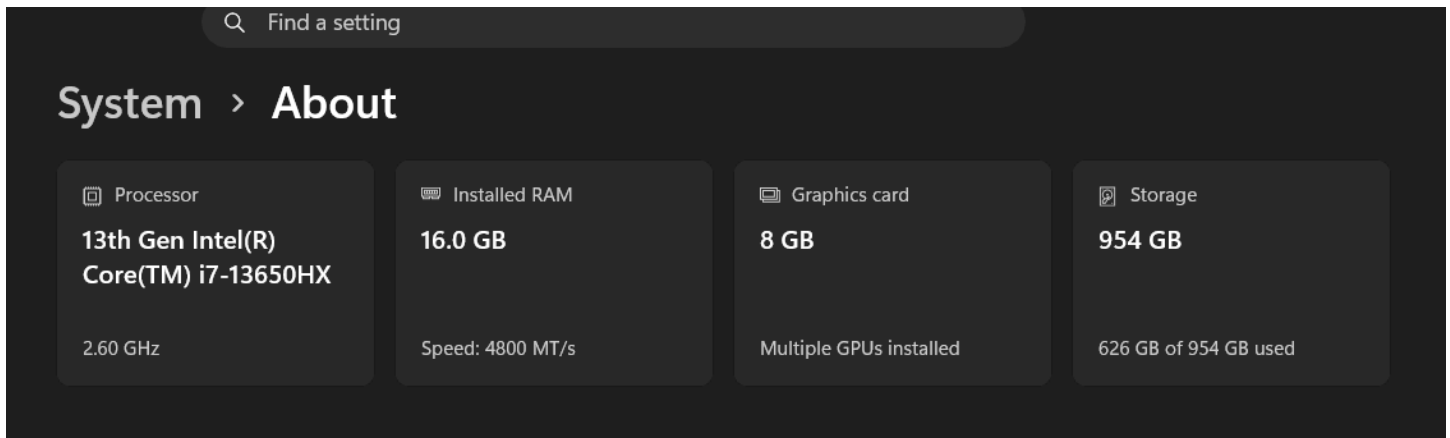


Working with large data sets in Power BI

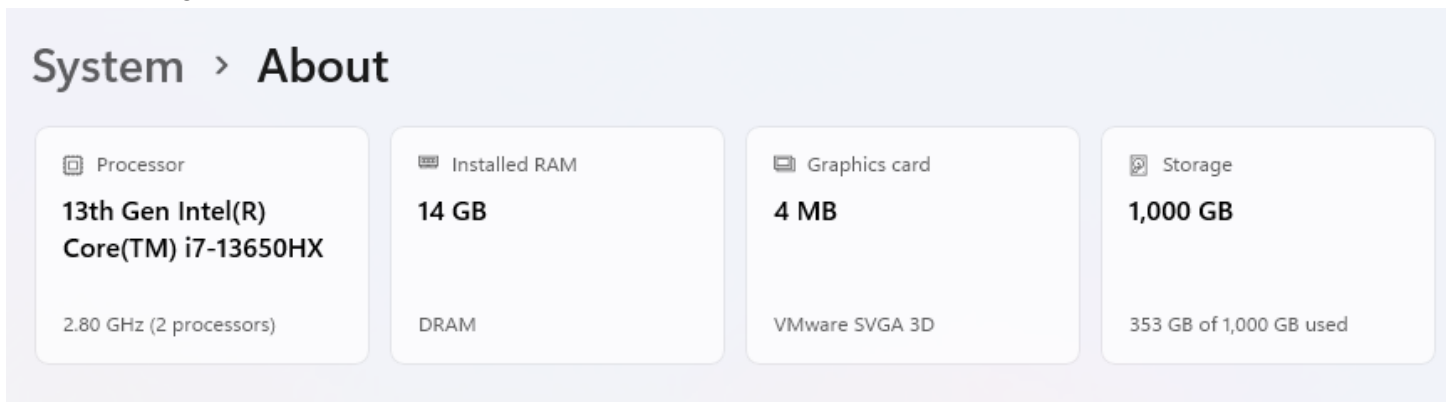
Host System Specification

The development machine is running on a Windows 11 host. The following snapshot shows the specification of the host machine.



The Virtual Machine Specification

The virtual machine is being run on an external hard drive. The specifications of the hosted system are shown in the following snapshot.



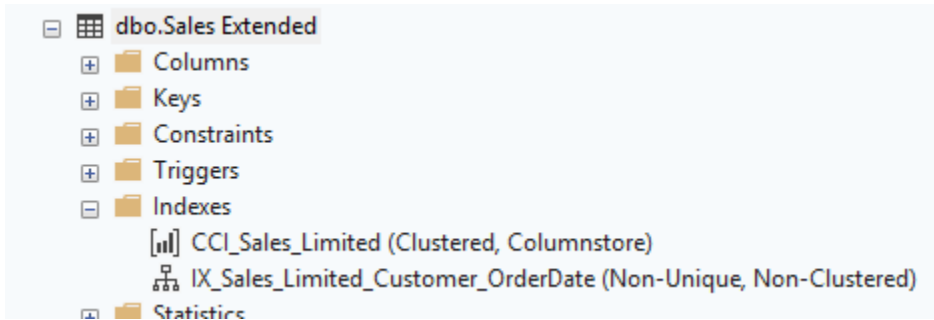
The Data

This Contoso database has the main Fact table of Sales. Following are snapshot shows the row count of the fact table.

A screenshot of a SQL Server query results window. The window has a toolbar at the top with a zoom dropdown set to 100%, a status bar showing 1 error and 0 warnings, and navigation arrows. Below the toolbar are tabs for 'Results' and 'Messages'. The 'Results' tab is active, showing a single row in a table with the title 'Row Count in Sales Table'. The row contains the value 237245485.

	Row Count in Sales Table
1	237245485

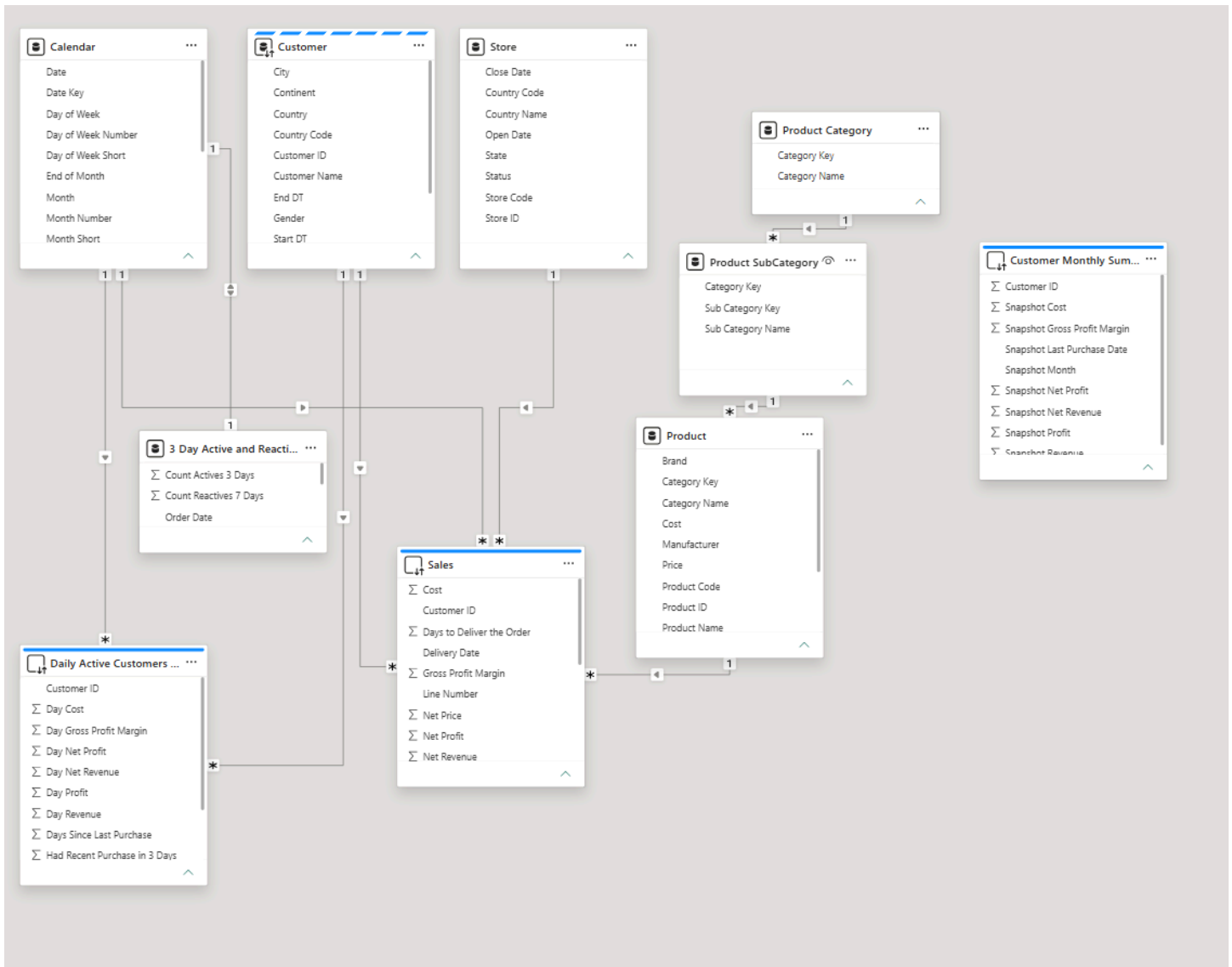
I have already created a clustered columnstore index on the table.



This speeds up the data processing on 237 million rows of data.

The Data Model in Power BI

Following snapshot shows the model of after several trials and errors.



Note that 3 tables are in Direct Query Mode. These tables are in Direct Query mode because of their size. They are stored in SQL Server Developer Edition 2025.

- Sales
- Daily Active Customers Trends
- Customer Monthly Summaries

Only one dimension table - Customer - is in Dual mode. This allows for the quick performance of the queries while the data itself is stored on SQL Server.

The Power Query Side

I have tried to organize the model in Power Query. Evidence is in the following snapshot.

The screenshot displays the Microsoft Power Query interface. The top ribbon includes tabs for File, Home, Transform, Add Column, View, Tools, and Help. The Home tab is active, showing options like Close & Apply, New Source, Recent Sources, Enter Data, Data source settings, Manage Parameters, Export query results, Refresh Preview, and Properties. Below the ribbon, the 'Queries [10]' pane on the left lists the following queries:

- Base tables [2]
 - Dimensions [6]
 - Calendar (selected)
 - Customer
 - Product
 - Store
 - Product SubCategory
 - Product Category
 - Facts [1]
 - Sales
 - Other Queries [3]
 - Customer Monthly Summaries
 - Daily Active Customers Trends
 - 3 Day Active and Reactive Customer Counts

The main area on the right shows a data table with the formula bar displaying `= Table.Rename`. The table has three columns: 'Date', 'AB', and 'C'. The data rows are numbered 1 through 35, with dates ranging from 1/01/2016 to 4/02/2016, and a constant value of 201 in column 'C'.

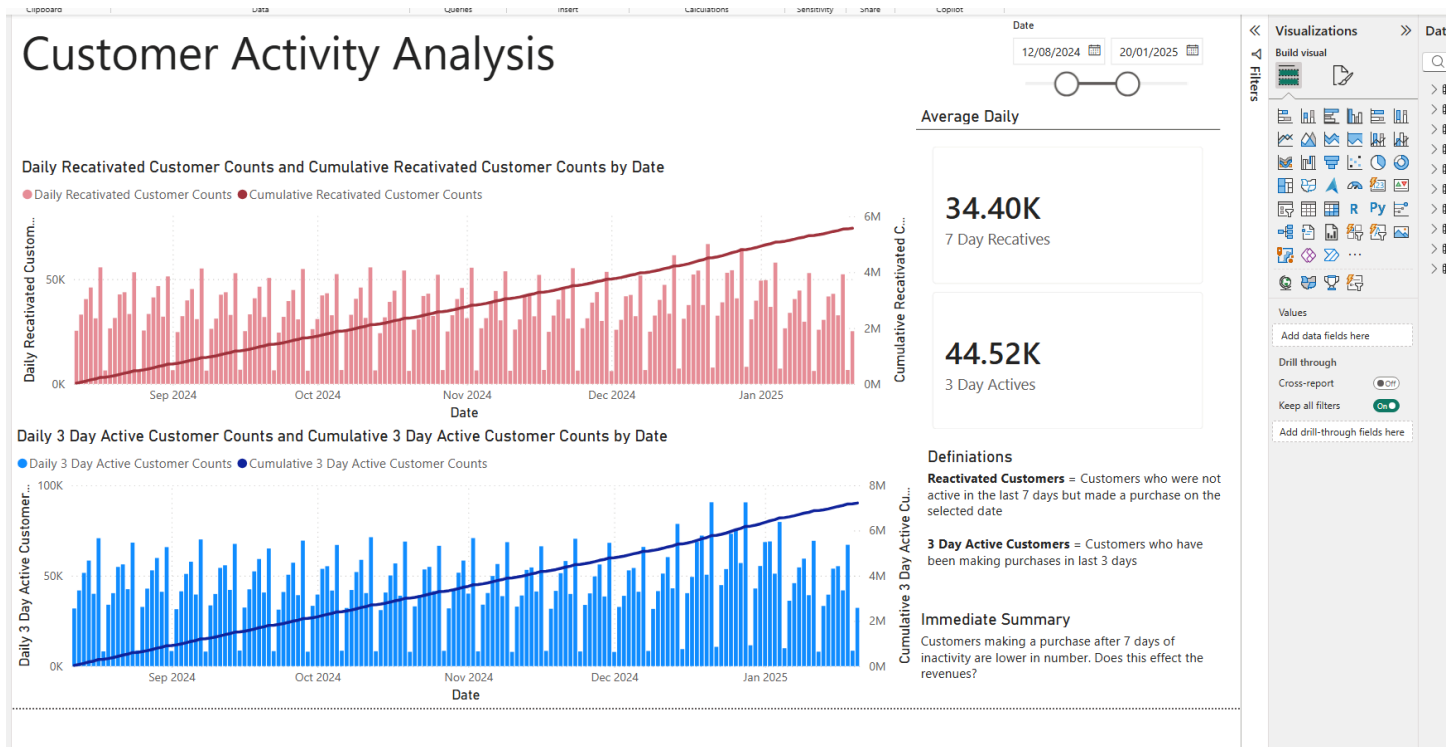
	Date	AB	C
1	1/01/2016		201
2	2/01/2016		201
3	3/01/2016		201
4	4/01/2016		201
5	5/01/2016		201
6	6/01/2016		201
7	7/01/2016		201
8	8/01/2016		201
9	9/01/2016		201
10	10/01/2016		201
11	11/01/2016		201
12	12/01/2016		201
13	13/01/2016		201
14	14/01/2016		201
15	15/01/2016		201
16	16/01/2016		201
17	17/01/2016		201
18	18/01/2016		201
19	19/01/2016		201
20	20/01/2016		201
21	21/01/2016		201
22	22/01/2016		201
23	23/01/2016		201
24	24/01/2016		201
25	25/01/2016		201
26	26/01/2016		201
27	27/01/2016		201
28	28/01/2016		201
29	29/01/2016		201
30	30/01/2016		201
31	31/01/2016		201
32	1/02/2016		201
33	2/02/2016		201
34	3/02/2016		201
35	4/02/2016		201

The Initial Draft of Power BI Report

The report focuses on customer categories based on the customer activities in two categories.

- Customers who were not active in last 7 days but are now active
- Customers who are active in the last 3 days.

The visuals show both the daily trends and cumulative counts based on the user selected slicer value.



Link to the file

https://github.com/faisaliqbal0x7ba/ContosoAnalysis_BigData